

Experiment 11

IP Spoofing using a GitHub Tool in Kali Linux

■ **Important Note:** This experiment is conducted strictly for academic and ethical purposes on authorized networks only. Unauthorized IP spoofing is illegal and punishable by law.

AIM

To demonstrate IP spoofing techniques using an open-source GitHub tool in Kali Linux for understanding network-level attack concepts and security vulnerabilities.

REQUIREMENTS

Host OS	Windows / Linux / macOS
Virtualization Software	Oracle VirtualBox
Guest OS	Kali Linux 2024
Tool Source	GitHub (IP Spoofing Tool — Educational Use)
Programming Language	Python / Bash (as per tool)
RAM	Minimum 4 GB (8 GB Recommended)
Disk Space	Minimum 30 GB
Processor	Intel/AMD with Virtualization support

THEORY

IP spoofing is a technique where an attacker forges the source IP address in network packets to impersonate another system. It is commonly used in attacks such as DDoS, session hijacking, and bypassing IP-based authentication. This experiment demonstrates IP spoofing using a publicly available GitHub tool to understand how attackers manipulate packet headers. Kali Linux provides a safe and controlled environment for ethical network security experimentation.

PROCEDURE

Step 1: Start Kali Linux Virtual Machine

- Open Oracle VirtualBox.
- Start Kali Linux.
- Log in using Kali credentials.

Step 2: Open Terminal

- Launch Terminal in Kali Linux.
- Update system packages: `sudo apt update`

Step 3: Clone IP Spoofing Tool from GitHub

- Clone the repository:
- `git clone https://github.com/<username>/<ip-spoofing-tool>`
- Navigate to the tool directory: `cd ip-spoofing-tool`

Step 4: Install Dependencies

- Install required dependencies:
- `sudo apt install python3`
- `pip3 install -r requirements.txt`

Step 5: Configure the Tool

- Open the script file: `nano spoof.py`
- Configure: Target IP address, Spoofed source IP address, Network interface

Step 6: Execute IP Spoofing

- Run the tool with root privileges:
- `sudo python3 spoof.py`
- Observe packet transmission with spoofed IP addresses.

Step 7: Monitor Network Traffic (Optional)

- Use Wireshark or tcpdump: `sudo tcpdump`
- Verify spoofed source IP packets.

COMMANDS

```
# Update system
sudo apt update

# Clone repository
git clone https://github.com//
cd ip-spoofing-tool

# Install dependencies
sudo apt install python3 -y
pip3 install -r requirements.txt

# Configure tool (edit spoof.py)
nano spoof.py

# Set: TARGET_IP, SPOOF_IP, INTERFACE

# Execute spoofing
sudo python3 spoof.py

# Monitor traffic (optional)
sudo tcpdump -i eth0 -n src
```

OUTPUT

Terminal Output

```
root@kali:~# sudo python3 spoof.py
[*] Starting IP Spoofing Tool...
[*] Interface : eth0
[*] Target IP : 192.168.1.10
[*] Spoofed IP : 10.0.0.99
[+] Sending 100 spoofed packets...
[+] Packets sent with forged source IP: 10.0.0.99
[+] Network traffic reflects spoofed IP information.
```

```
[+] Demonstration successfully completed.
```

```
# tcpdump verification
```

```
root@kali:~# sudo tcpdump -i eth0 -n
```

```
IP 10.0.0.99 > 192.168.1.10: ICMP echo request, id 1234
```

```
IP 10.0.0.99 > 192.168.1.10: ICMP echo request, id 1235
```

```
2 packets captured, 2 received
```

RESULT

IP spoofing was successfully demonstrated using a GitHub-based tool in Kali Linux. The tool forged the source IP address in outgoing packets and the spoofed IP (10.0.0.99) was confirmed in the tcpdump capture. The experiment illustrates how attackers can manipulate IP headers and highlights the need for robust network security mechanisms such as ingress filtering and packet authentication.